ENGINEERING SPECIFICATION

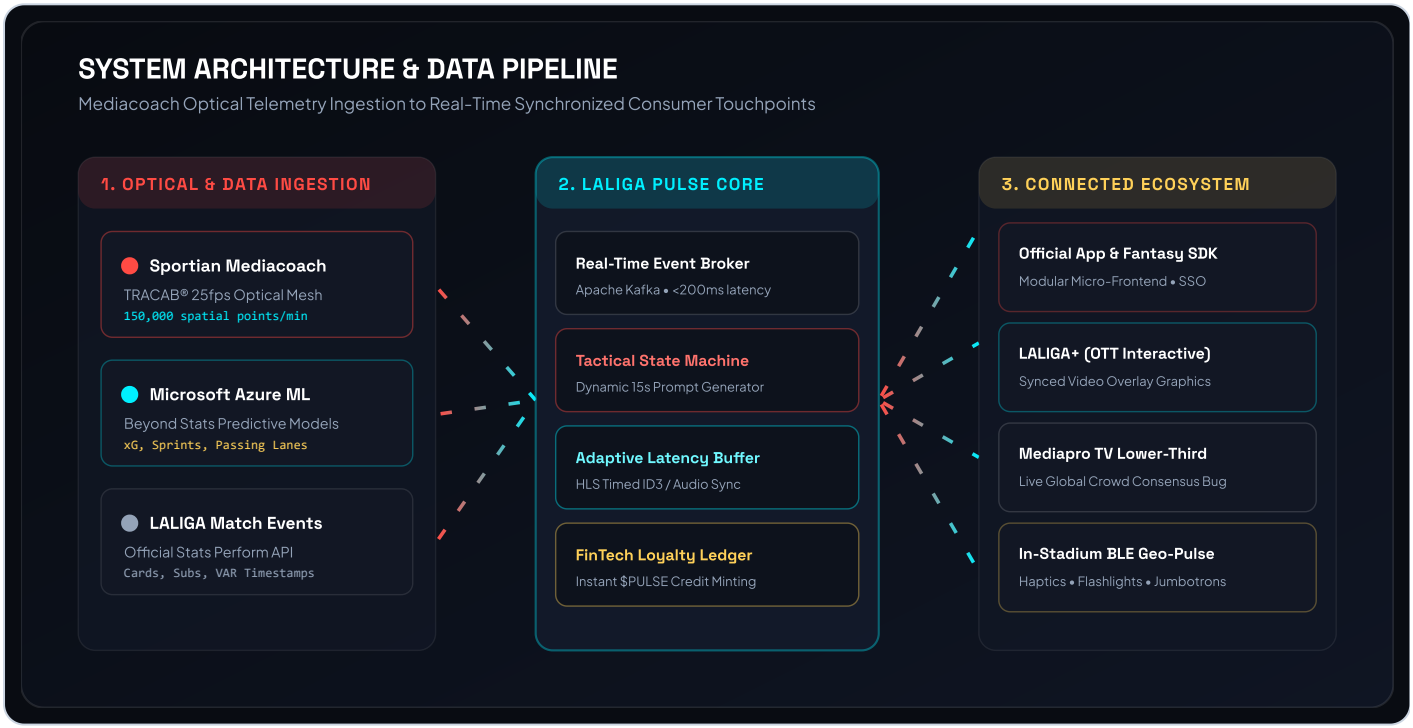
SPORTIAN & AZURE BEYOND STATS

LALIGA PULSE™ Technical Spec

Data Ingestion, Latency Synchronization, Event Contracts & Security Architecture

1. Optical Telemetry Ingestion (Sportian Mediacoach TRACAB)

The optical tracking array consists of 8 to 16 synchronized high-frame-rate cameras installed in the stadium catenary, capturing real-time spatial positioning of all 22 players and the ball at **25 frames per second**.



Field Name	Type	Frequency	Description
`match_id`	String (UUID)	Static	Official LALIGA fixture identifier
`frame_timestamp_utc`	Int64 (NTP ms)	25 Hz	Nanosecond-synchronized master timecode
`ball_coordinates`	Vec3 (X, Y, Z)	25 Hz	Ball 3D coordinates in normalized pitch space (0.0 to 105.0m)
`player_positions`	Array[PlayerPos]	25 Hz	22 objects with player_id, team_id, speed_kmh, and vector heading
`spatial_zone_index`	Int16 (1..18)	25 Hz	Current play zone (Zone 14: central edge of penalty box)

2. Adaptive Latency Synchronization Buffer

Global broadcast feeds suffer from physical delivery variance:

In-Stadium Spectators: **0.0s delay**
DTT / Terrestrial Antenna: **+3.5s delay**
IPTV / Cable: **+8.2s delay**
OTT Streaming (LALIGA+): **+18.4s delay**

To eliminate spoilers, the client SDK uses an **Adaptive Latency Sync Engine** that listens to an inaudible acoustic watermark or extracts HLS ID3 timed metadata tags from the user's video feed. Prompts appear synchronized to that specific fan's screen.

3. Apache Kafka Streaming & Partitioning Strategy

Kafka Topic	Partition Key	Retention	Consumers
`laliga.optical.tracking.raw`	`match_id` (16 Partitions)	24 Hours	Tactical State Machine, Spatial Aggregator
`laliga.tactix.prompts.active`	`challenge_id` (4 Partitions)	2 Hours	Redis Edge Sync, Push Notification Gateway
`laliga.fan.votes.stream`	`user_id_hash` (32 Partitions)	1 Hour	Consensus Aggregator, Audit Ledger
`laliga.settlement.outcomes`	`challenge_id` (4 Partitions)	7 Days	$PULSE Token Ledger, Broadcaster SDI Gateway

4. Data Protection & Security Boundary (GDPR & UAE PDPL)

LALIGA PULSE adheres to Privacy-by-Design principles. Fan identity is decoupled from behavioral telemetry using an ephemeral cryptographic token tied to LALIGA Single Sign-On (SSO). Geofencing is calculated on-device without continuous server tracking.